Spatial Context for Wildfire Ignition Prediction

Tre Pepper and Tanner Rohloff

The goal of this research-flavor project is to investigate whether wildfire ignition prediction in a 5 km x 5 km cell improves when collecting information from spatial context, and if these improved predictions justify additional computation cost.

Problem Statement

This Project is a collaboration between Tre Pepper and Tanner Rohloff. It is a Research-flavored project where we try to find whether conditions at multiple spatial scales improves wildfire ignition prediction compared to conditions available within only the local 5 km by 5 km cell. We model wildfire ignition risk using prior weather, drought, land cover, seasonal patterns, and historical fire activity. A local cell may omit the predictive information that a broader context will have, such as local vegetation, developed areas, and recent fire activity of the local area. We are trying to see if this broader context provides a predictive value to wildfire ignition across America. Each observation will be a grid-cell and prediction-date pair. The target will equal one if at least one qualifying wildfire in that grid cell during the following seven days, and zero otherwise. Qualifying fires will be restricted to fires greater than 0.5 acres after excluding recreational, unspecified, and other non-wildfire causes. The analysis completely depends on fire ignitions, not size, spread, severity, or intensity.

Project Goal and Motivation

Our goal is to build a temporal and spatial prediction pipeline using PostgreSQL/PostGIS and XGBoost. We are going to determine whether, and for which feature groups, broader context improves prediction beyond a local-only model. We are going to evaluate this empirically, as we do not know if performance will improve from this test. The prediction unit will remain a 5km x 5km cell in every experiment. We will compare the local only model with models that additionally incorporate information from surrounding 10, 20, and 50 km spatial cells. The 5km model will serve as the local baseline, while the larger-scale models will add neighborhood information surrounding the 5km prediction cell. In addition to predictive performance, we will record feature-generation and model-training costs for each spatial configuration. This will allow us to compare potential improvements with the additional compute and complexity of the models. Model outputs will include estimated risk probability and odds, and a current risk heatmap of the continental United States is a potential application.

Preliminary Methodology

We will assemble the datasets before training the models. All prediction observations will be represented by the common key (grid_id, prediction_date). Predictors will include temperature, daily precipitation, drought severity and history, NLCD land-cover percentages, seasonality, historical fire counts, and burned areas over short and multiyear windows. Weather will be assigned using the closest station with the required observations on the prediction date, and our features must use information available by that date. We will define the spatial grid before extracting features and will generate neighborhood summary statistics for the relevant spatial feature groups. The multi-scale experiments will use landcover and historical fire activity. Weather and precipitation will remain associated with the local

5km prediction cell because the current weather representations is station based rather than a continuous grid

Our evaluation is going to use non-overlapping chronological periods. We will be using 2000-2016 for training, 2017-2020 for validation, and 2021-2023 for testing. We will be using 2017-2020 to guide our model selection and parameter tuning, while our testing will remain held out until our final evaluations. Due to wildfire ignition being rare, we will be retaining all positives from training and sample around three negatives per positive. Validation and testing will use approximately five negatives per positive. Sampled probabilities will receive a prevalence adjustment and calibration assessment before being interpreted as real-world risk estimates. Our primary metric will be PR-AUC (Precision Recall- Area Under Curve), due to the rare nature of our target. Secondary metrics will include ROC-AUC (Receiver Operating Characteristic-Area Under Curve), precision, recall, brier score, F1, and log loss.

Milestones

Weeks 1–2: Audit the existing database, confirm source coverage and access, finalize neighborhood definitions, and validate target construction and temporal splits. **Weeks 3–4:** Extract local predictors, implement negative sampling, and train the 5 km baseline. **Weeks 5–6:** Build neighborhood features, then train and tune the four spatial configurations. **Week 7:** Complete held-out test evaluation, calibration checks, and ablation studies. **Week 8:** Consolidate results, document limitations and reproducible procedures, and prepare the final report and a heatmap prototype if feasible

Required Resources

The project requires wildfire records from the Fire Program Analysis Fire-Occurrence Database (FPA FOD), daily weather observations from NOAA/NCEI, U.S. Drought Monitor data, and annual NLCD land-cover data. Our software will include PostgreSql/PostGIS, Python, XGBoost, and data processing and evaluation libraries. PostgreSQL/PostGIS will be used to construct the spatial grid, store raw and processed relations, generate spatial neighborhoods, perform spatial and temporal joins, and produce the datasets used for ML. A multicore computer will be needed to support the computation for our initial experiments, however, a university computer, or computing resource may be needed. Expected limitations include the computational cost of grid-by-date and multiscale joins, as well as missing or imperfect source data, fire histories, and the temporal availability of land-cover data.

Workload Distribution

Tre Pepper will be responsible for data acquisition, database design, PostgreSQL/PostGIS implementation, data cleaning, spatial-temporal feature engineering, and database validation. Tanner Rohloff will be responsible for machine-learning implementation, XGBoost development, model tuning, and evaluation. Both team members will contribute to experimental design, interpretation of results, the final report, and the final presentation.

Related Works:

This paper servs as a relevant example of machine learning prediction for igniting wildfires using spatial and environmental data. The focus on spatial structure and the probability of ignition lays a basis for the exploration of how various levels of spatial context influence ignition predictions.

https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025EF006935